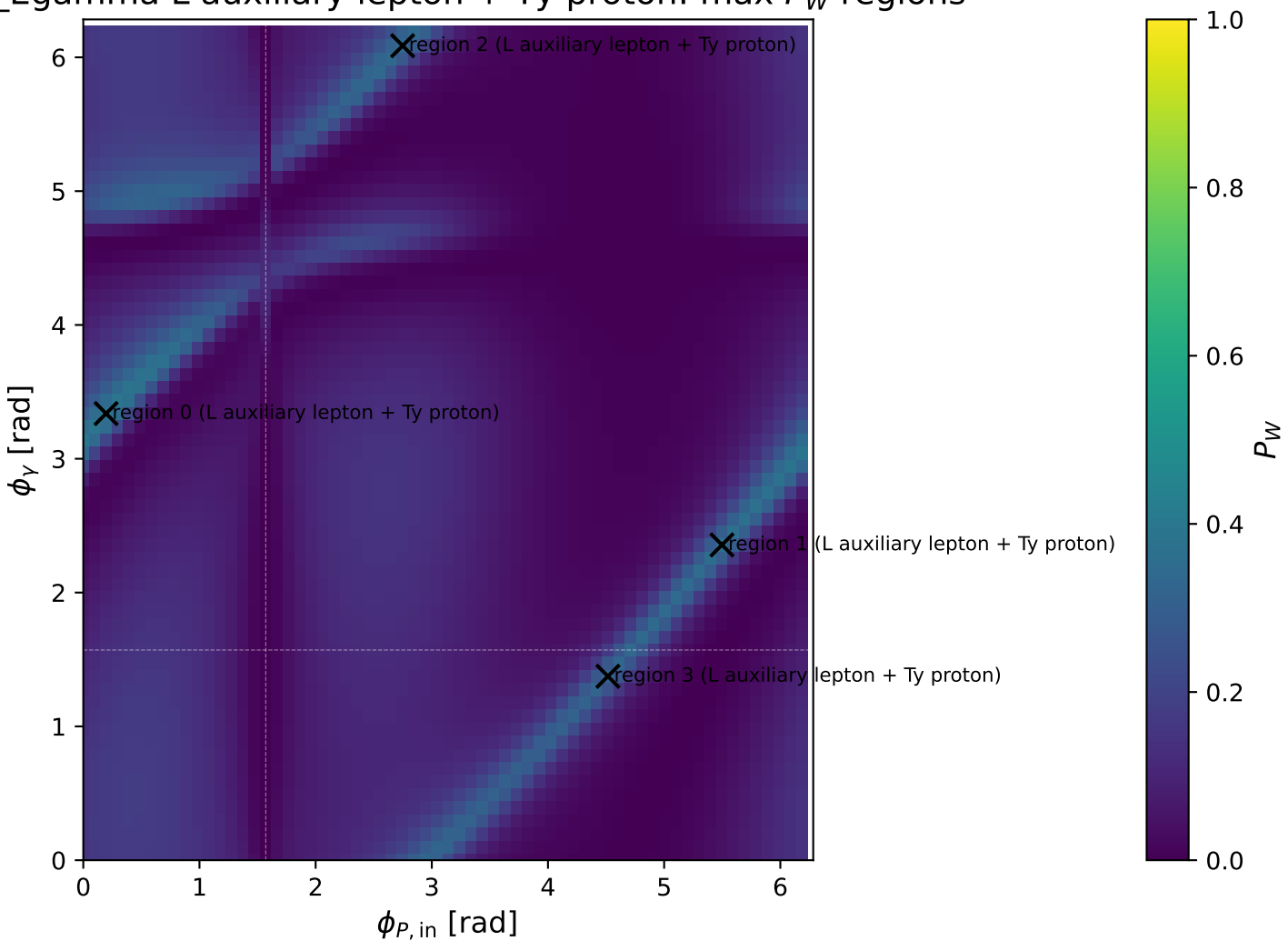
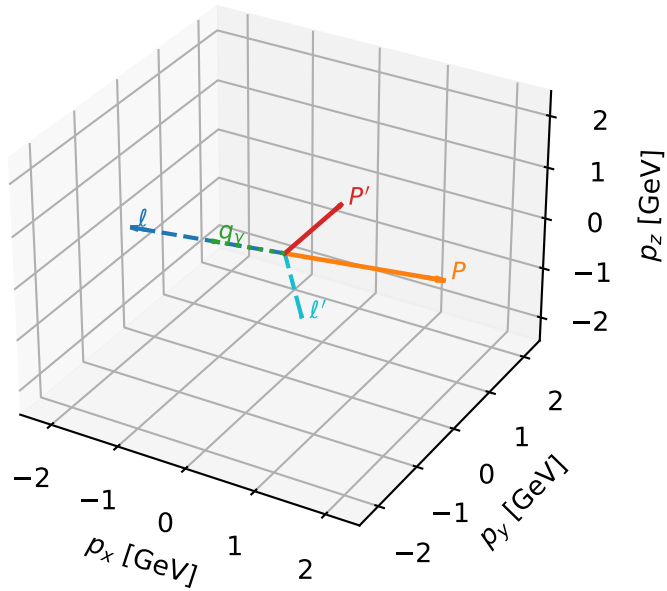


Egamma L auxiliary lepton + Ty proton: max P_W regions



L auxiliary lepton + Ty proton characteristic max P_W configuration

3D momenta

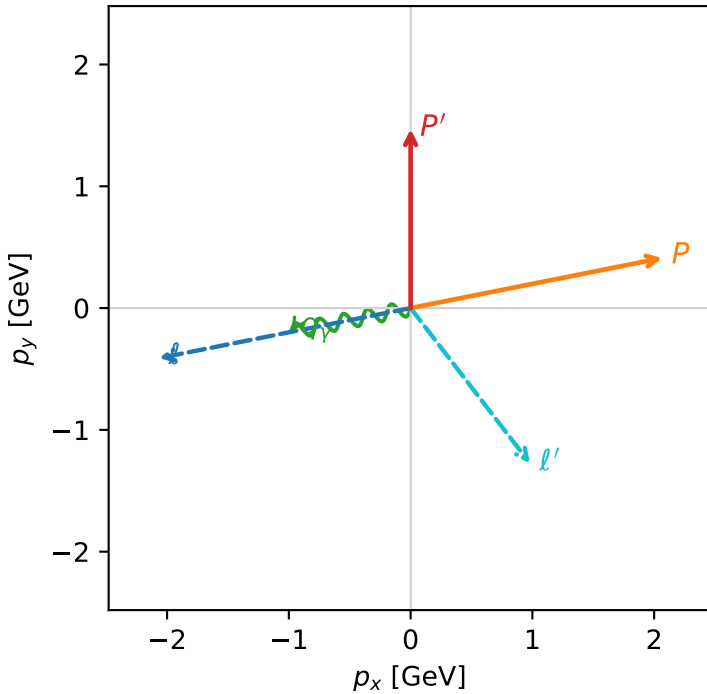


w_purity_mid_egamma_lty_region_0 P_W=0.390427
 region: mid_Egamma_LTy_region_0
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|L_l\rangle \otimes |T_{y,p}\rangle$

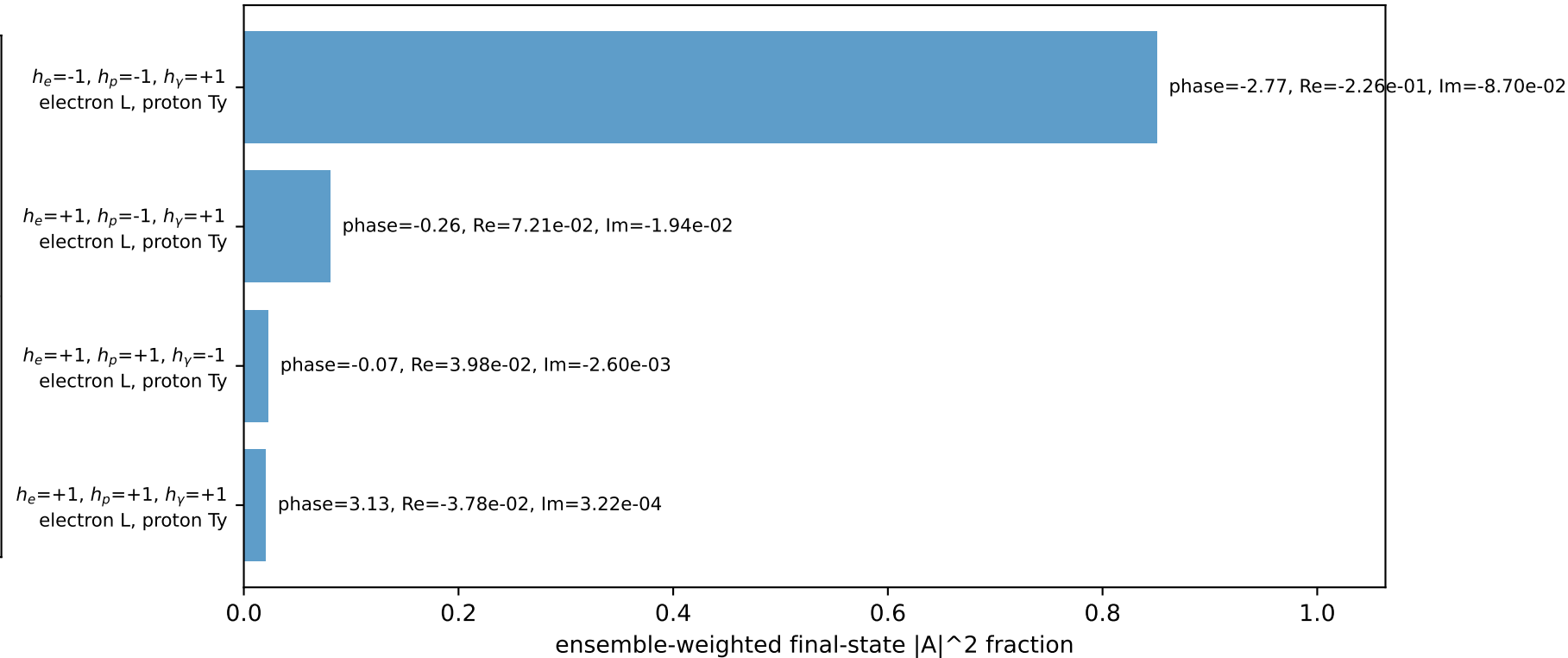
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.47047$
 $\theta_{in}=1.57079$, $\phi_l=3.33794$, $\phi_P=0.19635$
 $E_Y=1$, $\phi_Y=3.33794$
 $Q^2=9.84105$, $x_B=0.707829$, $t=-5.07665$, $W^2=4.94194$, $y=0.708044$
 $\theta(l', \gamma)=116.311$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.3322$ $p_x= -2.0665$ $p_y= -0.4110$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= 2.0665$ $p_y= 0.4110$ $p_z= 0.0000$
 l' $E= 1.8943$ $p_x= 0.9808$ $p_y= -1.2754$ $p_z= 0.0000$
 P' $E= 1.7442$ $p_x= 0.0000$ $p_y= 1.4705$ $p_z= 0.0000$
 q_Y $E= 1.0000$ $p_x= -0.9808$ $p_y= -0.1951$ $p_z= 0.0000$

Transverse plane

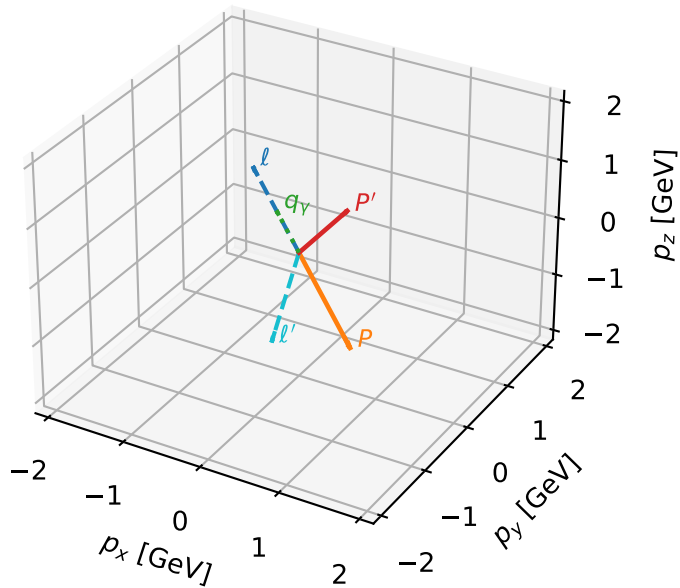


Leading initial-ensemble/final-state components (N=4, retained=97.5%)



L auxiliary lepton + Ty proton characteristic max P_W configuration

3D momenta



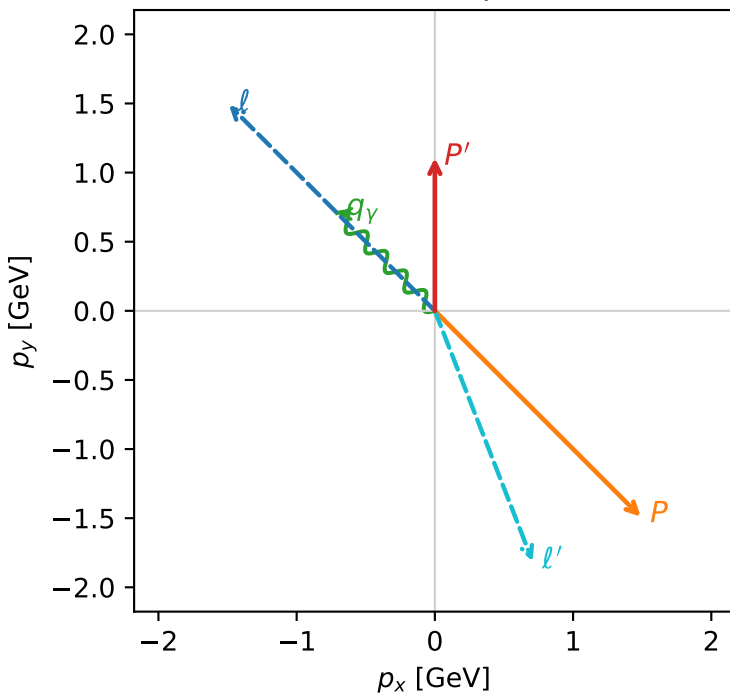
w_purity_mid_egamma_lty_region_1 P_W=0.356066
 region: mid_Egamma_LTy_region_1
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|L_\ell\rangle \otimes |T_{y,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.1062$
 $\theta_{in}=1.57079$, $\phi_\ell=2.35619$, $\phi_p=5.49779$
 $E_\gamma=1$, $\phi_\gamma=2.35619$
 $Q^2=15.7165$, $x_B=0.921637$, $t=-8.22632$, $W^2=2.21615$, $y=0.868447$
 $\theta(\ell', \gamma)=156.304$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:

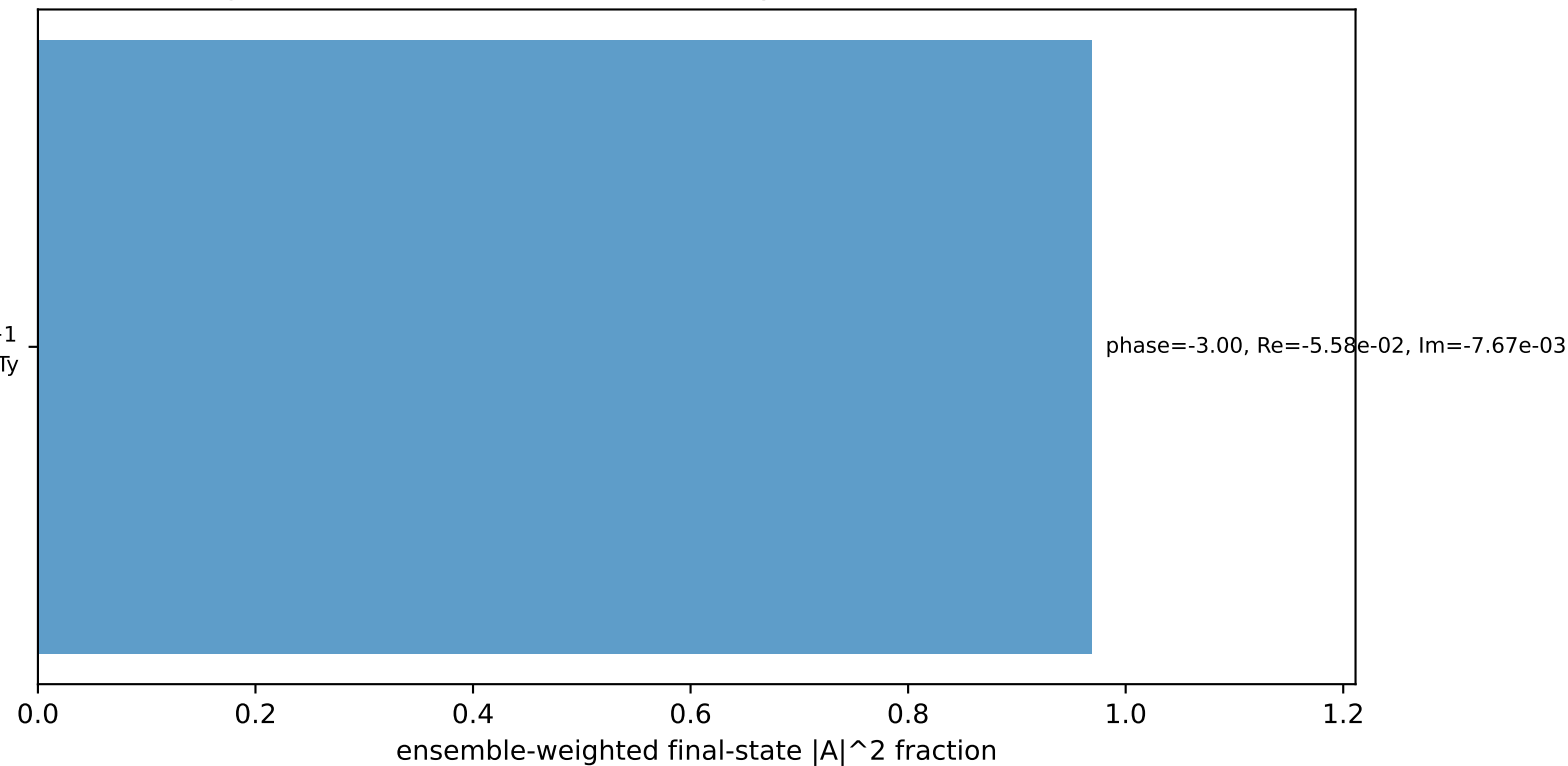
ℓ $E= 2.3322$ $p_x= -1.4898$ $p_y= 1.4898$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= 1.4898$ $p_y= -1.4898$ $p_z= 0.0000$
 ℓ' $E= 2.1882$ $p_x= 0.7071$ $p_y= -1.8133$ $p_z= 0.0000$
 P' $E= 1.4504$ $p_x= 0.0000$ $p_y= 1.1062$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= -0.7071$ $p_y= 0.7071$ $p_z= 0.0000$

Transverse plane



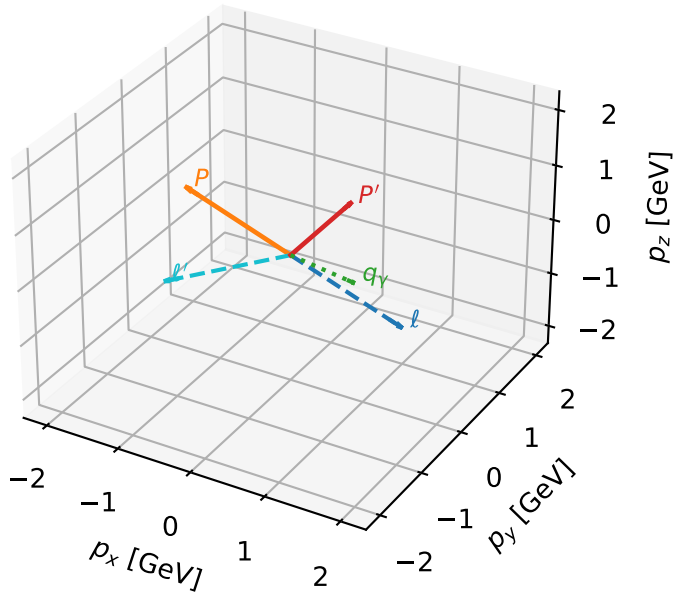
$h_e=-1$, $h_p=-1$, $h_\gamma=+1$
 electron L, proton Ty

Leading initial-ensemble/final-state components (N=1, retained=96.9%)



L auxiliary lepton + Ty proton characteristic max P_W configuration

3D momenta



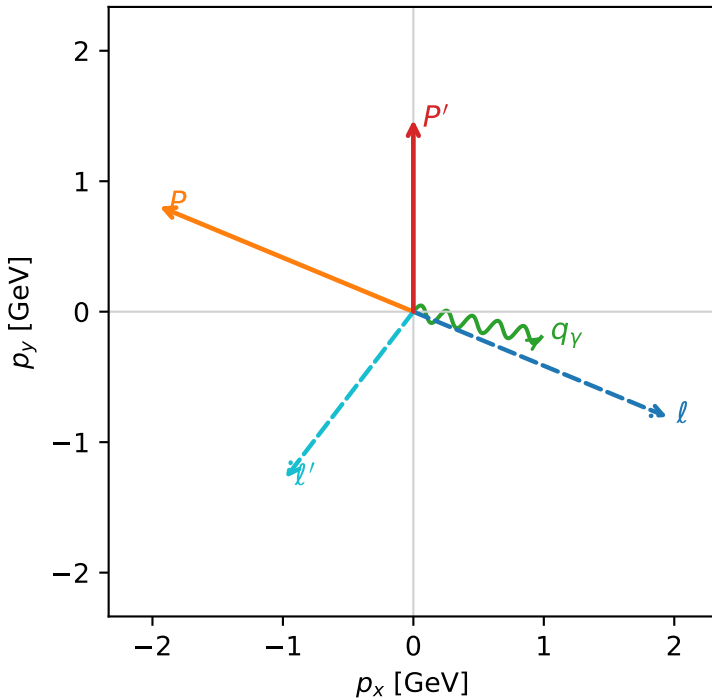
w_purity_mid_egamma_lty_region_2 P_W=0.338917
 region: mid_Egamma LTy_region_2
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|L_l\rangle \otimes |T_{y,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.47047$
 $\theta_{in}=1.57079$, $\phi_l=5.89049$, $\phi_p=2.74889$
 $E_\gamma=1$, $\phi_\gamma=6.08684$
 $Q^2=8.59768$, $x_B=0.679134$, $t=-3.91424$, $W^2=4.94194$, $y=0.644723$
 $\theta(l', \gamma)=116.311$ deg

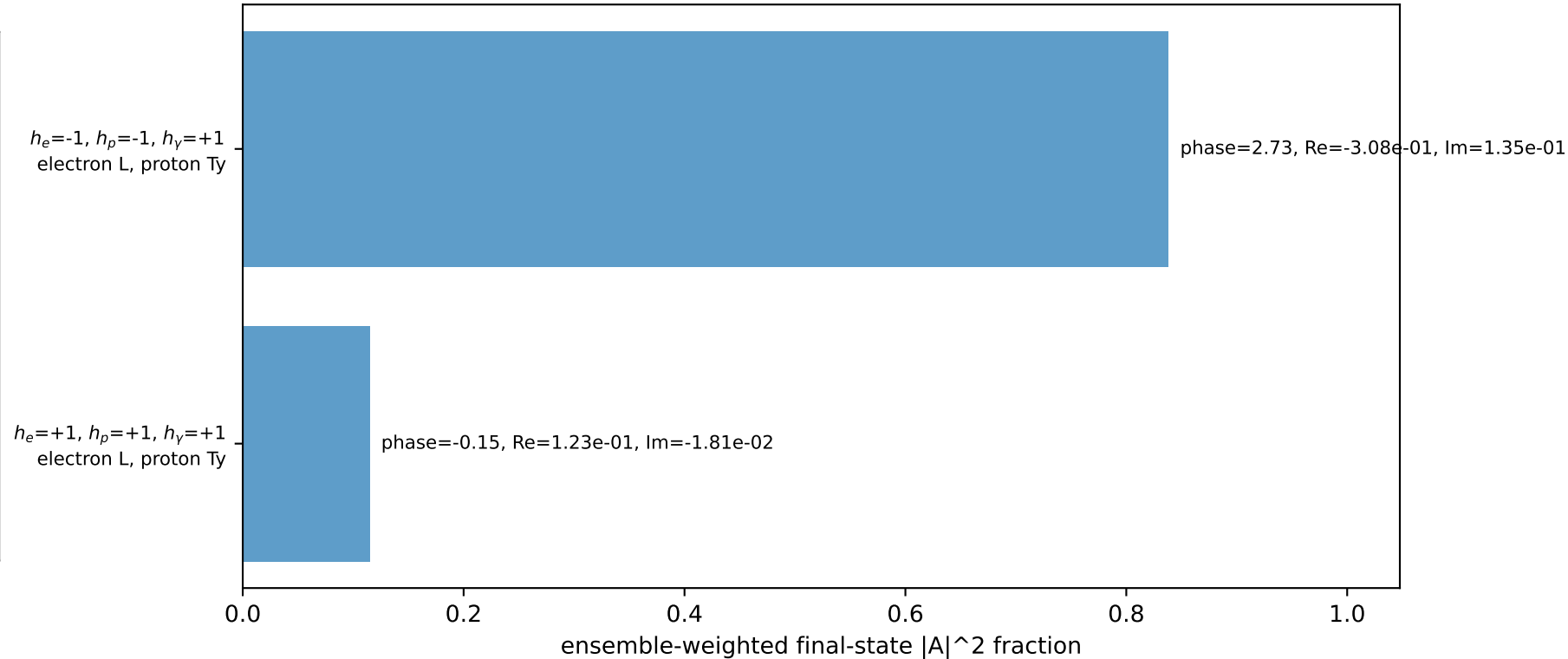
four-momenta $[E, p_x, p_y, p_z]$ GeV:

l $E= 2.3322$ $p_x= 1.9466$ $p_y= -0.8063$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= -1.9466$ $p_y= 0.8063$ $p_z= 0.0000$
 l' $E= 1.8943$ $p_x= -0.9808$ $p_y= -1.2754$ $p_z= 0.0000$
 P' $E= 1.7442$ $p_x= 0.0000$ $p_y= 1.4705$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= 0.9808$ $p_y= -0.1951$ $p_z= 0.0000$

Transverse plane

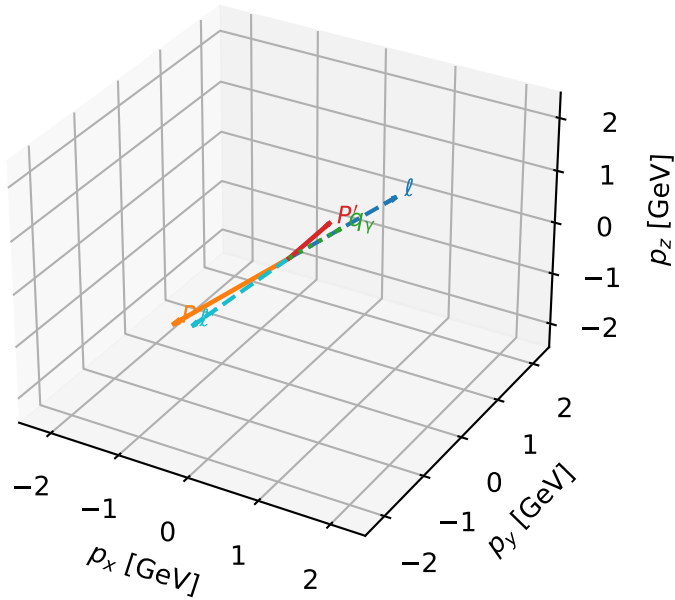


Leading initial-ensemble/final-state components (N=2, retained=95.3%)



L auxiliary lepton + Ty proton characteristic max P_W configuration

3D momenta

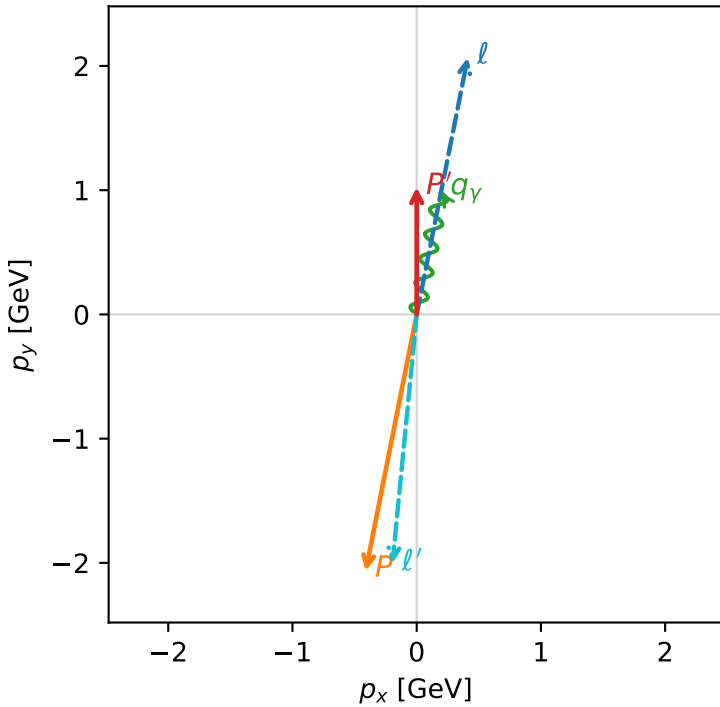


w_purity_mid_egamma_lty_region_3 P_W=0.325577
 region: mid_Egamma_LTy_region_3
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|L_\ell\rangle \otimes |T_{Y,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.02462$
 $\theta_{in}=1.57079$, $\phi_\ell=1.37445$, $\phi_P=4.51604$
 $E_Y=1$, $\phi_Y=1.37445$
 $Q^2=16.9406$, $x_B=0.95661$, $t=-8.88253$, $W^2=1.64824$, $y=0.901866$
 $\theta(\ell', \gamma)=174.306$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3322$ $p_x= 0.4110$ $p_y= 2.0665$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= -0.4110$ $p_y= -2.0665$ $p_z= 0.0000$
 ℓ' $E= 2.2494$ $p_x= -0.1951$ $p_y= -2.0054$ $p_z= 0.0000$
 P' $E= 1.3891$ $p_x= 0.0000$ $p_y= 1.0246$ $p_z= 0.0000$
 q_Y $E= 1.0000$ $p_x= 0.1951$ $p_y= 0.9808$ $p_z= 0.0000$

Transverse plane



$h_e=-1$, $h_p=-1$, $h_Y=+1$
 electron L, proton Ty

Leading initial-ensemble/final-state components (N=1, retained=99.5%)

